

Response to Reviewers – CDIO 2026 (Paper ID 139)

Summary

I thank the reviewers for their constructive and detailed feedback. The revisions focus on strengthening the evidence base, clarifying the assessment-theory positioning, improving operational clarity, and making the scope and limitations of the approach more explicit, while maintaining the design-oriented nature of the contribution.

Reviewer 1

1. Strengthen assessment-theory positioning

Response:

The manuscript now explicitly connects the proposed assessment architecture to formative and process-oriented assessment literature. In particular, PR-based workflows are framed as mechanisms that make feedback, revision, and learning processes visible and assessable. This connection is articulated in the section *Assessment Focus: Reproducibility, Handover, and Responsiveness*, linking observable behaviors to established assessment principles (Biggs & Tang, 2015).

2. Clarify the evidence base

Response (addressed):

Lightweight quantitative evidence has been added in the *Student Response and Adaptive Refinement* section. Specifically:

- Trends in rubric-based assessment scores (GitHub usage and teamwork)
- Statistical indicators (slopes and p-values)
- Observed changes in PR size, documentation quality, and review depth

These provide indicative, non-experimental evidence aligned with the paper's design-oriented scope.

3. Elaborate limitations and boundary conditions

Response:

A dedicated "Boundary conditions and limitations" paragraph has been added (following the design principles section), addressing:

- Student readiness and need for scaffolding
 - Workflow overhead in early stages
 - Scalability considerations
 - Transferability to non-technical domains
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4. Provide additional operational clarity

Response:

The definition of *substantive peer review* has been clarified in the assessment section, specifying:

- Actionable, content-relevant feedback
- Engagement in follow-up discussion
- Distinction from superficial approvals

This improves reproducibility for other instructors.

Reviewer 2

1. Literature review sufficiency

Response:

The literature positioning has been refined to:

- Explicitly acknowledge the broader use of GitHub in education
- Clarify the gap regarding PR-based workflows as primary assessment artifacts

This strengthens positioning without expanding the review beyond scope.

2. Quantifiable measures vs previous methods

Response:

The manuscript now includes:

- Quantitative trends from rubric-based data

While no controlled experiment is conducted, the paper now provides indicative comparative evidence.

3. Limitations of the method

Response:

Limitations are now explicitly addressed in the “Boundary conditions and limitations” section, including:

- Overhead
 - Scalability
 - Transferability constraints
 - Dependence on prior training
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4. PR concept clarity (addressed)

A concise definition of PRs has been added early in the paper.

5. Course level and enrollment (addressed)

Course level, size, and team structure are now explicitly stated.

Reviewer 3

1. Lack of empirical evidence

Response (addressed):

Quantitative and observational evidence has been added, including:

- Rubric-based trends
- Behavioral changes across course iterations

The manuscript clearly positions these as indicative rather than experimental findings.

2. Lack of visual representation (addressed)

A workflow diagram (Figure 1) has been included to clarify interaction structure.

3. Link between CDIO skills and measurement

Response (addressed):

The manuscript now explicitly maps:

- Reproducibility → documentation practices
- Handover → PR descriptions
- Responsiveness → review engagement

These are tied to observable repository evidence and rubric-based evaluation.

Final remarks

The revised manuscript maintains its design-oriented contribution while improving:

- empirical grounding (within scope),
- theoretical positioning,
- clarity of implementation,
- and transferability.

I believe the revisions fully address the reviewers’ concerns while preserving the paper’s core contribution.